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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

7287040288

ENVIRONMENTAL MANAGEMENT **5014/01**

Paper 1 **May/June 2008**

2 hours 15 minutes

Candidates answer on the Question Paper.

Additional Materials: Ruler

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Section A

1 **(a)** In September 2006 a newspaper report stated:

PROTESTS IN BANGLADESH OVER LARGE OPENCAST MINE PLAN

The people protesting claim that the opencast mine will displace 40 000 villagers from their homes and land and ruin their livelihoods just for 1 000 new jobs. They believe that wells within 6 km of the mine will dry up and rivers will be polluted. The mining company promises these advantages: new jobs, foreign exchange, tax revenues for the government and new infrastructure.

State **one** social disadvantage and **two** economic advantages that will result from the opening of this mine. Explain each.

social disadvantage .....

explanation .....

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economic advantage 1 .....

explanation .....

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economic advantage 2 .....

explanation .....

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.....[6]

(b) After mining has finished in an area, how can the damaged environment be improved?

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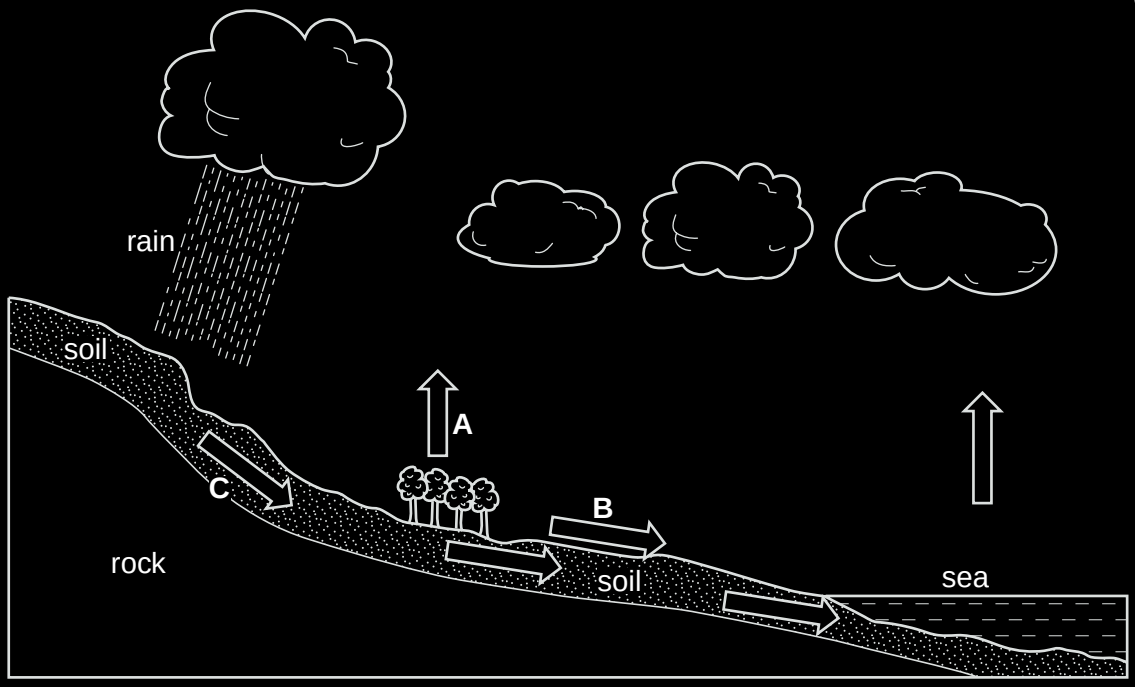
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2 **(a)** The diagram shows a water cycle.



(i) Name the processes shown by the arrows.

A .....

B .....

C .....[3]

(ii) Explain how water from the sea may return to the sea.

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.....[4]

(b) Suggest why some rural areas of the developing world are suffering from a shortage of groundwater from wells and boreholes.

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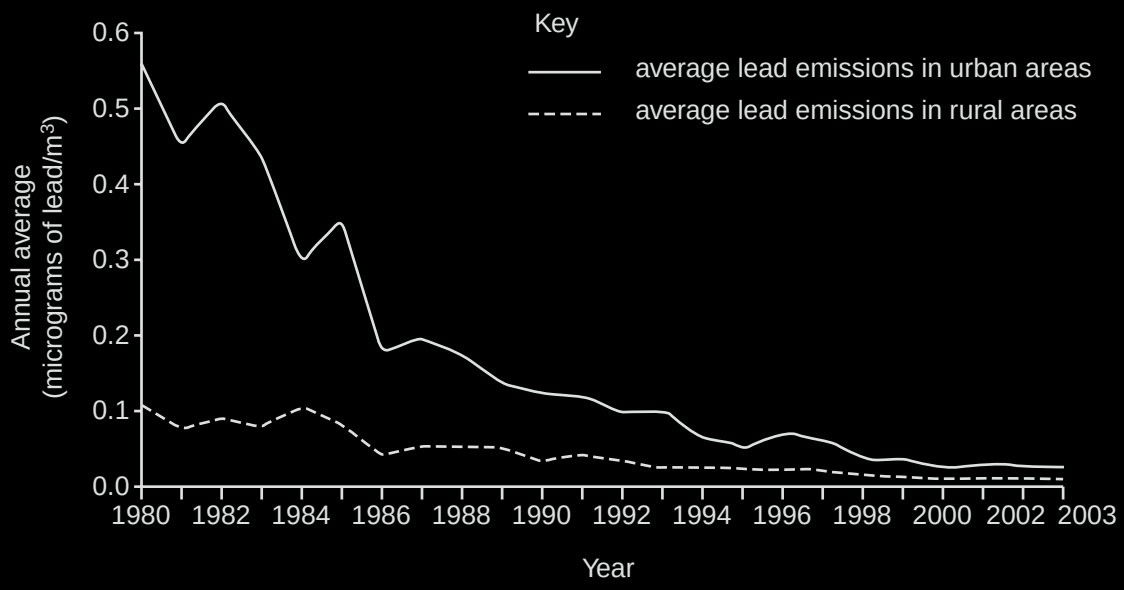
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3 **(a)** Look at the graph showing information about lead emissions in the UK.



(i) Describe the main trends in lead emissions **in urban areas** shown in the graph between 1980 and 2003.

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.....[3]

(ii) Suggest reasons for the trends shown in urban areas.

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(iii) Suggest why the amount of lead emitted in rural areas is different from that in urban areas.

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(b) Why is reducing lead emissions important?

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.....[2]

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4 **(a)** The photograph shows a deforested area that is now used for grazing sheep. Som
ago, a small part of it was enclosed by fences to investigate vegetational successio



(i) Look at the photograph. What evidence is there that vegetational succession is taking place inside the fence?

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.....[2]

(ii) What type of plant cover is likely if the succession continues to its climax?

.....[1]

(iii) Why does animal grazing stop vegetational succession?

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.....[1]

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(b) How does a vegetational succession improve the soil over time?

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(c) The area shown in the photograph has high rainfall throughout the year. The present ecosystem is poorer than that of the original forest. Why?

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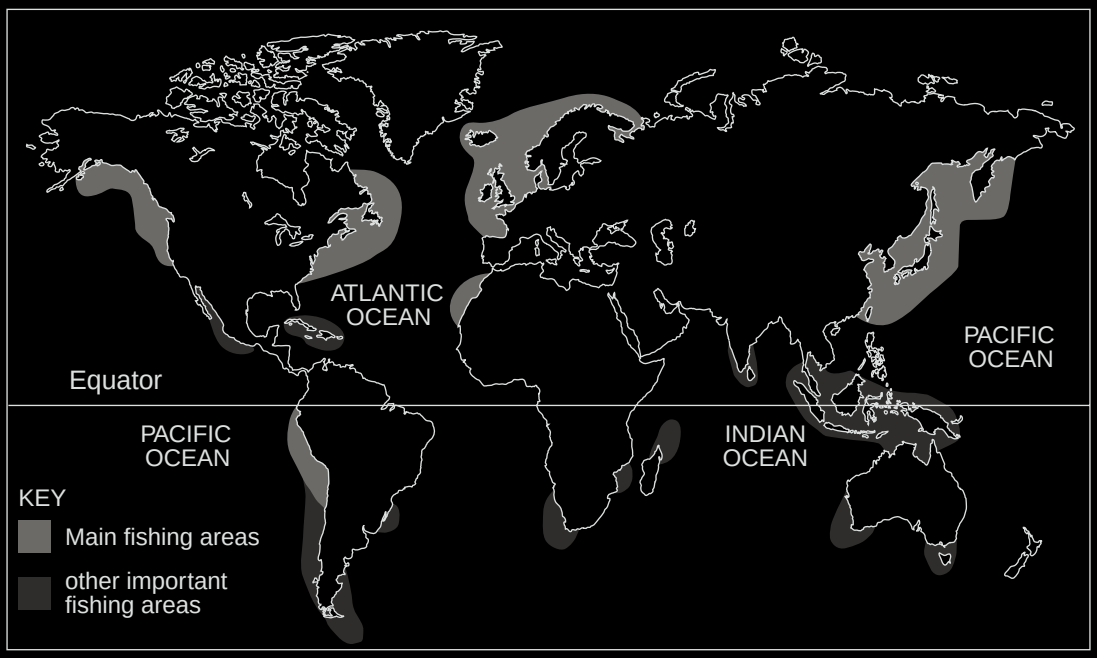
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Section B

5 **(a)** Look at the world map showing the main and other important ocean fishing areas.

World Ocean Fisheries



(i) Describe where the **main** fishing areas are located.

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.....[3]

(ii) State one similarity and one difference between the locations of the main and other important fishing areas.

Similarity .....

.....

Difference .....

.....[2]

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(b) Three factors to explain the locations of major ocean fisheries are listed below.

- wide continental shelf
- presence of ocean currents
- densely populated coastal areas nearby

(i) How many of these factors are natural (physical) factors?

.....[1]

(ii) Choose **two** of the factors. For each one, explain its importance for major ocean fisheries.

Name of factor .....

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Name of factor .....

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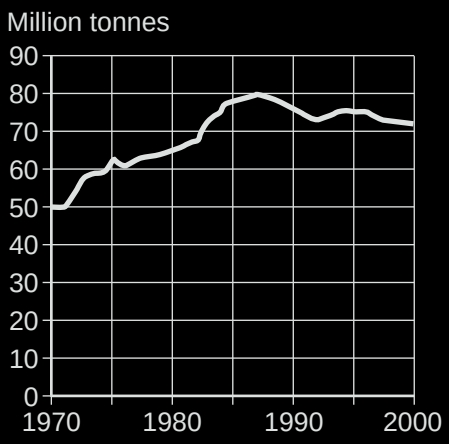
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(c) Look at the graph showing total world ocean fish catches.

World Ocean Fish Catches



(i) State the amount of fish caught in 1970, 1987 and 2000.

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.....[2]

(ii) Describe the evidence from the graph which suggests overfishing.

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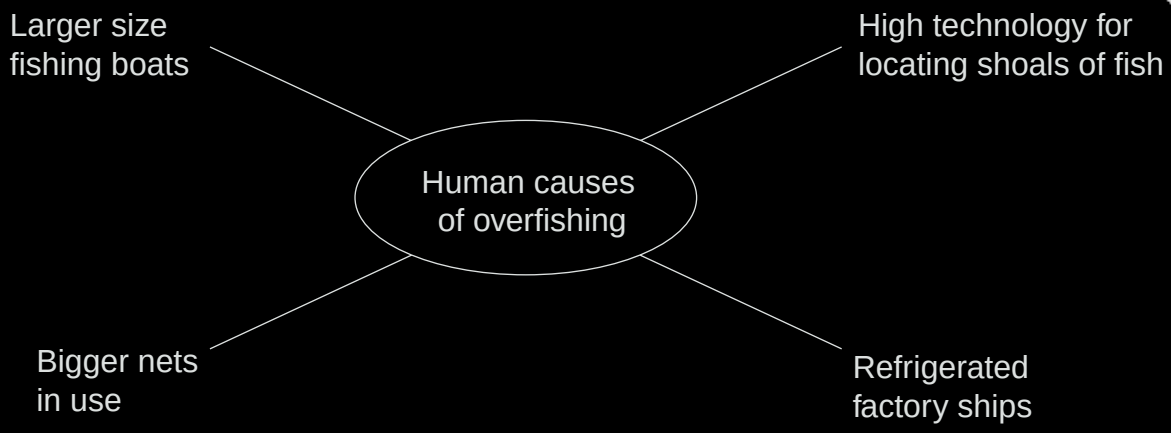
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(d) Some of the human causes of overfishing are shown in the spider diagram.



(i) Briefly describe how each one contributes to overfishing.

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(ii) In your view, which one is contributing most to overfishing? Explain why.

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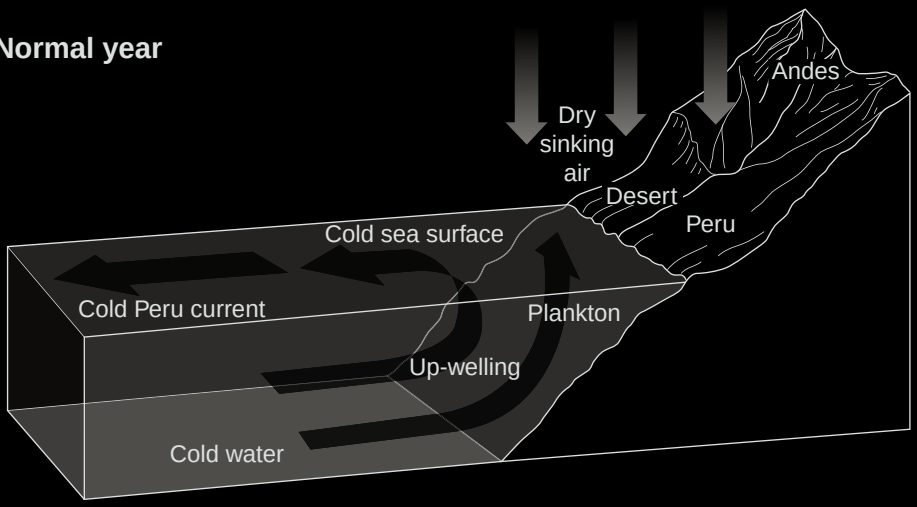
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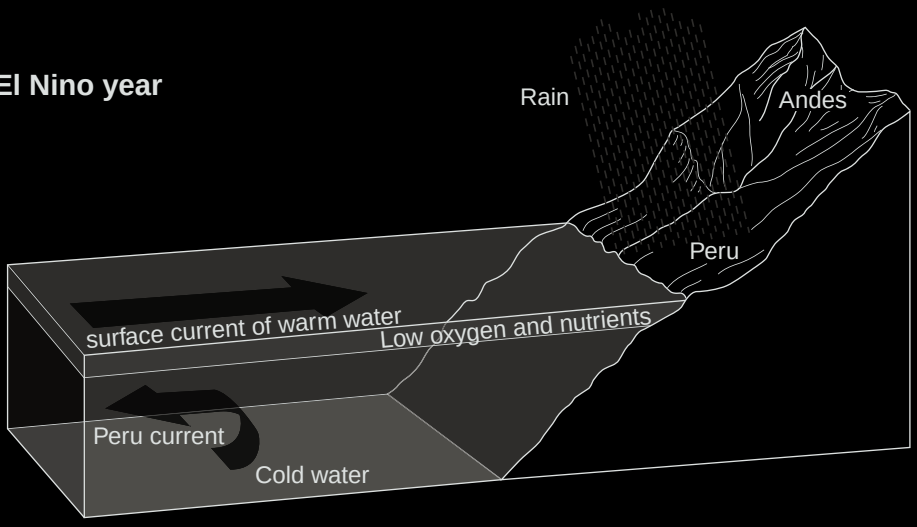
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(e) In some ocean fishing areas, fish catches are reduced by natural causes. This h
off the coast of Peru in El Nino years.

Normal year



El Nino year



Describe the differences between normal and El Nino years for

(i) ocean currents

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(ii) warm and cold water

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.....[2]

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(iii) Explain why there are fewer fish off the coast of Peru in El Nino years.

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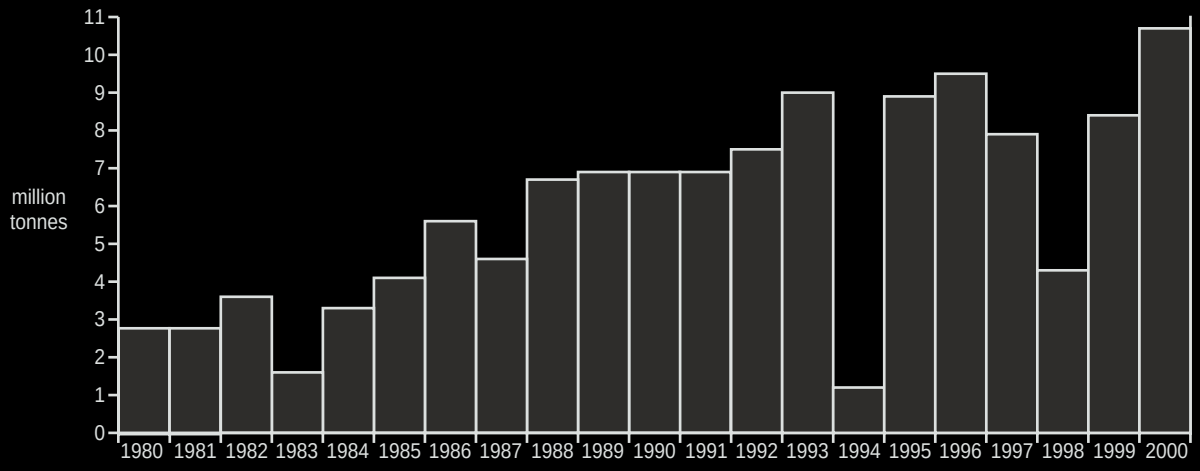
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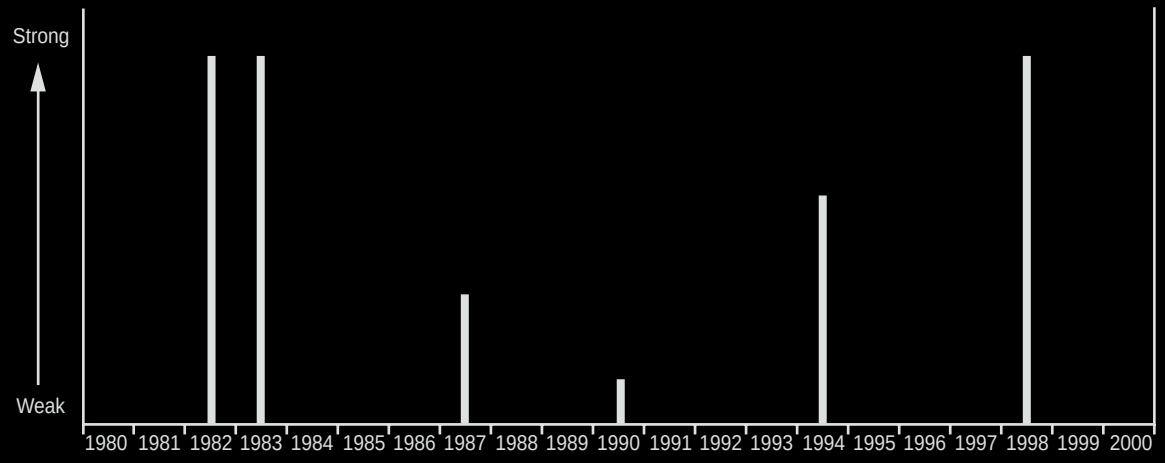
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(f) Look at the graph for fish catches in Peru and the graph for El Nino years.

Total fish catches in Peru



The record of El Nino years in Peru



(i) State the three strong El Nino years.

.....[1]

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(ii) Do El Nino years have an effect on the size of fish catches in Peru? Suppo answer with evidence from the graphs.

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.....[4]

(iii) Is there any evidence, from the graph of fish catches, for overfishing in Peru during the 1990s? Explain your answer.

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.....[2]

(g) (i) Draw another spider diagram to show three strategies for the sustainable harvesting of ocean fisheries and reducing overfishing.

[4]

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(ii) Are these strategies for the sustainable harvesting of ocean fisheries e… difficult to carry out? Explain your views on this.

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.....[5]

[Total: 40]

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6 **(a)** Look at the photo of a weather station.



(i) State why this is a good place to site a weather station.

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.....[2]

(ii) Give one reason why a fence is usually built around weather stations.

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.....[1]

(iii) Name a weather instrument that will be placed inside the white wooden box.

.....[1]

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(iv) Draw a labelled diagram to show how wind speed is measured in a w station.

[3]

(b) Look at the table of climate data for two weather stations in Africa north of the Equator.

Station A

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature (°C)	16	19	23	28	31	34	34	34	33	30	24	19
Precipitation (mm)	0	0	0	0	0	0	0	0	0	0	0	0

Station B

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature (°C)	22	24	28	31	29	29	27	25	26	27	25	21
Precipitation (mm)	0	0	3	10	69	117	206	310	142	13	0	0

(i) Complete the table below for Station A.

	Station A	Station B
Highest monthly temperature (°C)		31
Lowest monthly temperature (°C)		21
Annual range of temperature (°C)		10
Precipitation during the year		wet and dry season

[3]

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(ii) When is the wet season at Station B?

.....[1]

(iii) Major climatic types in Africa

Equatorial Savanna Desert

Name the type of climate at Station A and at Station B.

Station A .....

Station B .....[1]

(iv) State one reason for each of the choices made in part **(iii)**.

Station A .....

.....

Station B .....

.....[2]

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(c) Many parts of Africa are too dry for cultivation for part or all of the year. Four options to allow farming in dry areas are listed below.

1 Practise extensive livestock farming
2 Plant new varieties of crops
3 Government builds large dams, reservoirs and irrigation canals
4 Use underground water supplies for trickle drip irrigation

(i) For three of the options, state one advantage and one disadvantage of each for farming in dry areas.

Option number .....

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Option number .....

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Option number .....

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(ii) In your view, which one of the four options is the most sustainable for farming in dry areas? Explain your choice.

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.....[3]

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(d) **(i)** Name **one** area where desertification is a major problem.

.....[1]

(ii) There are both physical and human causes of desertification.
Explain this statement.

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.....[3]

(iii) Which is more important as a cause of desertification – human or physical factors?
Explain your view.

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.....[2]

(e) In big cities located in dry climates, such as Los Angeles, atmospheric pollution is a big problem.

(i) State **two** reasons why atmospheric pollution from traffic and industry is often worse in places with a dry climate.

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(ii) **Los Angeles**

Number of days with atmospheric pollution worse than recommended health standards

Year	Number of days
1975	210
1980	170
1985	160
1990	130
1995	95
2000	45

In the space below, draw a graph to show these values.

[4]

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(iii) Describe the trend from 1975 to 2000.

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(iv) Where are strategies for reducing traffic emissions likely to be more effective – in developed world cities like Los Angeles, or in developing world cities like Cairo and Beijing?

Explain your answer as fully as you can.

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.....[4]

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